MAE502 Project Report

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Project Repository

F1 Challenge Final Report

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A. ABSTRACT

This report details the control methodology and trajectory optimization for a rear-wheel-drive (RWD) front-wheel-steer F1 car racing through the Circuit of America (COA). The system utilizes a bicycle model constrained by real-world track boundaries and absolute tire force limits. By decoupling the path generation from the discrete control loop, a Software-in-the-Loop (SIL) method was deployed to test digital PID and LQR controllers against a continuous open-loop simulation. This project highlights tradeoffs in controller tuning for vehicles, as well as the challenge of mapping between continuous and discrete simulation environments. Ultimately, the system achieved a zero-violation closed-loop lap time of 178.30 seconds. The open-loop verification provided critical insights into the limitations of Zero-Order Hold (ZOH) integration within continuous ODE solvers.

B. TRAJECTORY GENERATION & OPTIMIZATION

The foundation of the vehicle's trajectory optimization relies on the maximization of each corner's inner radius in the COA. By mapping the corresponding "s" values of the track every 250 meters, an iterative "out-in-out" racing line was established. The vehicle swings wide on the corner entry, cuts in and hugs the corner's apex, and then swings wide once again on the exit. This effectively maximizes the corner radius, reducing the vehicle's lateral force demand and allowing for more longitudinal force output, keeping the car's speed generally higher.

Based on the track width and the ISO coordinate system, lateral offsets were assigned as $\pm$ 4.0 meters. This offset magnitude establishes a distance from the track's centerline that does not put the car on the edge of the track, while also allowing it to swing out enough to maximize the corner radius. On the straights of the track, the targeted offset returns to the track's centerline (0 meters). To map these waypoints into a continuous trajectory, they were spliced together via the "makima" command, which widens the apexes without any mathematical overshoot. The resulting path trajectory is seen below in Figure 1.

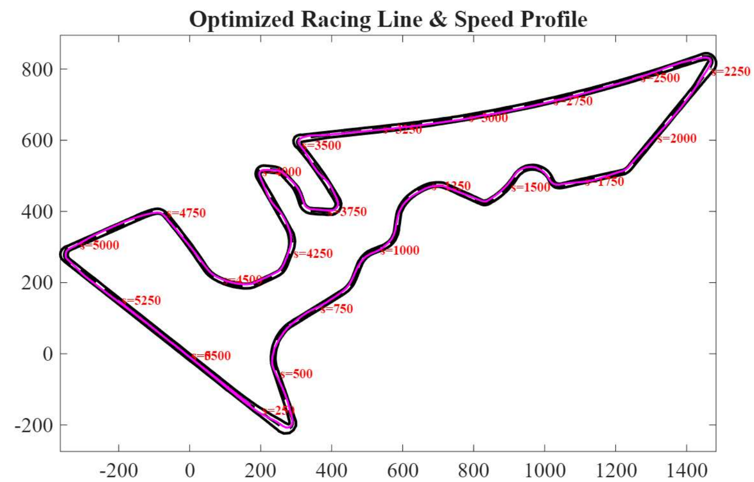

Figure 1: Optimized COA Trajectory

To manage the dynamic tire loading throughout the lap, the path generation script creates a velocity profile based on manual speed constraint sections and a global speed multiplier. By increasing the global speed scale, the aggression of the lap would increase and push the vehicle to its limits for a better lap time. The manual sections act as a primary "tuning knob" that helps to eliminate lateral force violations found during the closed-loop simulations. When a violation was identified, its corresponding s-location would be found, and a 20-meter velocity reduction zone was implemented around the apex of the force violation in order to prompt the vehicle to decelerate earlier

To smooth out the acceleration and deceleration demands of the car, forward and backward sweeps through the trajectory's indices were implemented. These sweeps consider the elliptical tire mode that defines the tradeoffs between max lateral and longitudinal force based on the tire's capabilities. To provide more "available force" for lateral movement, the max longitudinal acceleration and deceleration were capped to 95% of their actual maximum. This forced the trajectory planner to initiate earlier and more gentle braking zones that allow for larger lateral forces that allow the vehicle to corner without any violations. The resulting velocity profile is seen below in Figure 2.

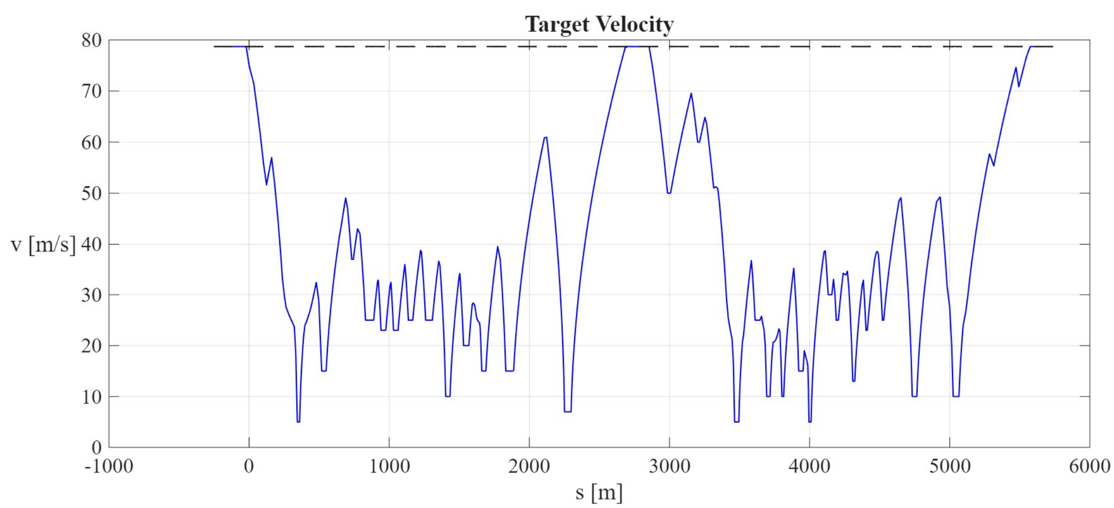

Figure 2: Path Generation Target Velocity Profile

While the project's overarching goal is robust vehicle control, the path generation phase proved to be very important in highlighting an aspect of vehicle dynamics; feedback control cannot make up for a physically infeasible or suboptimal trajectory. By thoroughly developing the geometric path optimization and velocity profile that helps to manage erratic actuation demands, the path generation file provides a strong foundation for the closed-loop controller to operate at a much higher efficiency and yield an incredibly low lap time.

C. CLOSED-LOOP CONTROL ARCHITECTURE

The control strategy of this project comprises two digital controllers: a PID longitudinal speed controller as well as an LQR lateral steering controller. The choice to push the controllers into the discrete-time domain was in attempts to directly match the sampling frequency of the simulation. To execute this within a continuous ODE solver, both control functions utilize persistent variable memory. Whereas typical functions clear their local variables between each call, the persistent memory locks the previous states within the controller and allows it to successfully mimic a digital Engine Control Unit (ECU). The controller only updates its command every 0.05 seconds and holds the calculated Zero-Order Hold (ZOH) efforts constant across the solver's sub-steps. This memory retention is crucial for both controllers' performance, as without it, the accumulated integral error, derivatives, and elapsed time would be erased between sub-steps and therefore the controllers' tracking capabilities would be completely disabled.

The longitudinal controller takes inspiration the provided exemplary "baseline" speed control (a PI controller) supported with a derivative gain and an aerodynamic feedforward term to help eliminate phase-lag in heavy braking sections. The feedforward term finds the target acceleration directly from the path file and then outputs the necessary longitudinal force that will achieve such an acceleration (if possible) before the PID calculates the error. The derivative gain on the PID then acts as a preemptive damper to keep the car from overshooting the target speed. This helps to slow the car down efficiently while also freeing up grip for the lateral requirements of cornering. As can be seen in Figure 3, the PID follows the desired velocity with high precision, where the only major deviation occurs on the main straightaway at s = [2500 3000] meters, when the target velocity outpaces the vehicle model's acceleration limits. Ironically, the physical saturation acts as a safety limit, providing the vehicle with enough braking distance for the upcoming heavy braking section.

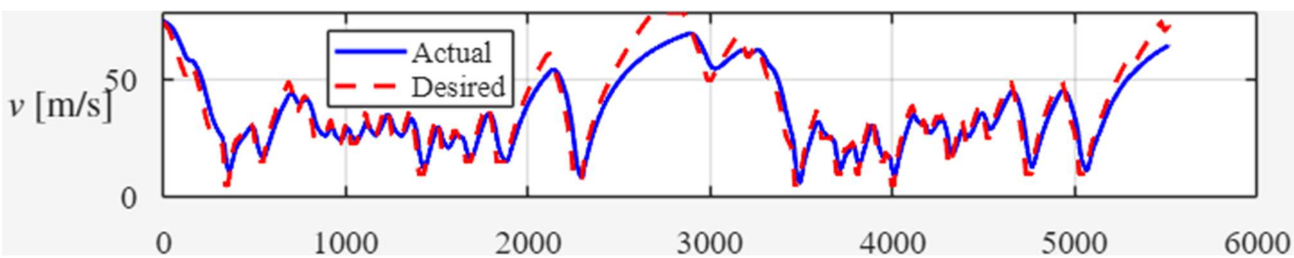

Figure 3: Actual vs. Desired Velocity Profile

The LQR steering controller is composed of a highly penalized control effort (R) with a moderate tolerance of both tracking and heading errors (Q_{11} & Q_{22}, respectively). For some reference of the scale, the control effort penalty weight is 550 times that of the penalty on either error. This combination allows for smooth steering inputs that avoid excessive lateral force spikes that cause violations in the simulation. Additionally, a proportional gain was added to further ensure the physical steering commands are executed quickly to reach the desired values without any phase lag. As can be seen in Figure 4, the LQR performs very well in achieving the desired γ values with its $\dot{\gamma}$ inputs.

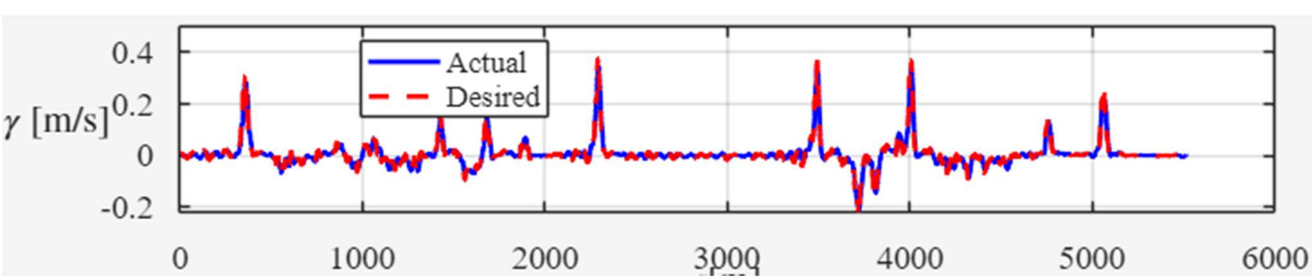

Figure 4: Actual vs Desired γ Profile

During the development phase, an Iterative Learning Control (ILC) algorithm was implemented as an outer loop to further reduce tracking and speed error within the controllers across several simulated laps. While the ILC succeeded in its objective, the result was a lap time that increased. This led to the removal of the ILC and highlighted that because the generated trajectory is already somewhat conservative to avoid force violations, forcing the vehicle to be more precise in its tracking was slower than allowing for some drift in the controllers. This means that in these simulations, reducing localized error does not necessarily result in improved global performance.

D. OPEN-LOOP VERIFICATION FAILURE DISCUSSION

Following the closed-loop validation of the 178.30 second lap, the control arrays (R & $\dot{\gamma}$) were saved as inputs that would further validate the control strategy and viability in an open-loop environment. Despite receiving the exact optimized inputs, the open-loop simulation saw an extreme divergence in integrated terms leading to a large number of lateral force and track violations. A root-cause analysis ended up revealing that the failure was not a result of poor controller tuning or physics modeling, but a data truncation error within the simulation's Runge-Kutta 4th Order (RK4) continuous ODE solver.

The discrepancy between the two simulation environments is derived from how the "CarSimRealTime" function handles the active LQR feedback loop during numerical integration. Inside the closed-loop architecture, the active controller recalculates state derivatives at every RK4 sub-step, finding micro-adjustments during the midpoint sub-steps. However, the master extraction array (U) only logs the control value generated at the initial k_1 sub-step. This means that about 75% of the digital ECU's stabilizing calculations are truncated and permanently lost in the input extraction.

When the open-loop simulation tries to reconstruct the lap, it is relying on linear interpolation (interp1) to guess those missing mid-step control values. This estimation fails to capture the dynamic adjustments that the closed-loop steering controller utilized to prevent tire saturation. This data truncation is seen directly in the vehicle's physical telemetry within the first fraction of a second, as displayed below in Table 1.

Table 1: Integration Divergence at RK4 Sub-Step

Time (t)	CL Input ($\dot{\gamma}$)	OL Input ($\dot{\gamma}$)	CL State (γ)	OL State (γ)	CL Position (X)	OL Position (X)
0.15 s	0.011529	0.011529	0.004679	0.004679	3.8786	3.8786
0.20 s	0.011529	0.011529	0.005255	0.005255	6.7964	6.7964
0.25s	-0.012426	-0.012426	0.004834	0.005632	9.7213	9.7220
0.30 s	-0.024895	-0.024895	0.003693	0.005011	12.6507	12.6546

As can be seen, the closed-loop and open-loop simulations remain identical through 0.20 seconds. However, exactly at 0.25 seconds, despite identical interpolated control inputs, the lack of an active mid-step LQR calculation causes the steering state to diverge. In the closed-loop system, the LQR immediately notices any position error and corrects it. However, since the open-loop system does not utilize a feedback mechanism, it blindly integrates a flawed state, resulting in an even further deviated state. This error compounds exponentially, leading to the vehicle's performance to continuously worsen throughout the lap. Very quickly, the vehicle sees massive tire saturation, goes off-track, and consequently fails in the open-loop simulation. This can very obviously be seen in Figure 5 below.

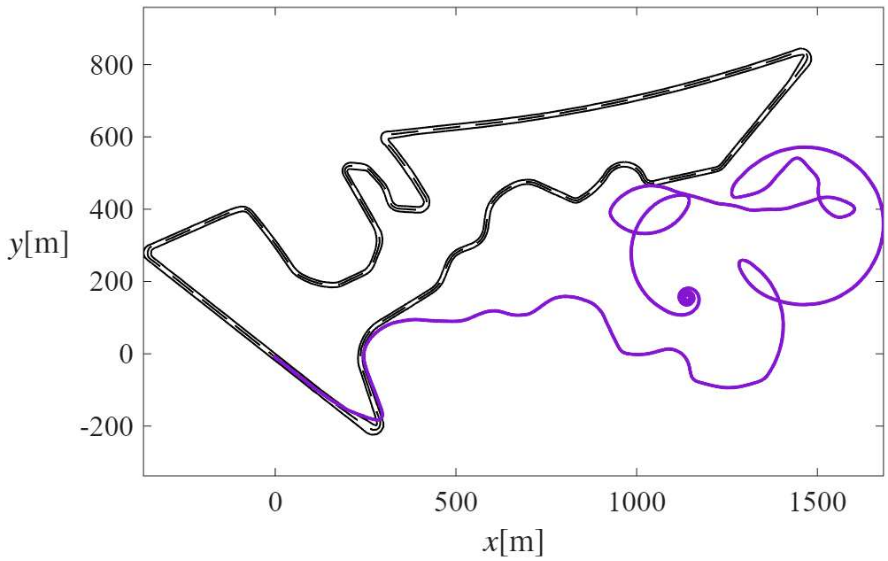

Figure 5: Open-Loop Verification Failure

E. RESULTS & CONCLUSION

The F1 challenge project brought about many engineering insights regarding trajectory optimization and discrete controller tuning to push a system's performance to its physical limits. Through the development of a friction-circle conscious velocity profile along with robust PID speed and LQR steering control, the system was able to achieve a closed-loop lap time of 178.30 seconds. The system did so without exceeding saturation limits regarding longitudinal force, lateral force, and steering rate. To verify the vehicle's performance within these restrictions, Figures 6 and 7 display the actual control inputs and resulting tire forces plotted alongside their respective boundaries. As can be seen, the combined tire forces remain strictly within the 5000 and 6000 N friction limits defined within the project.

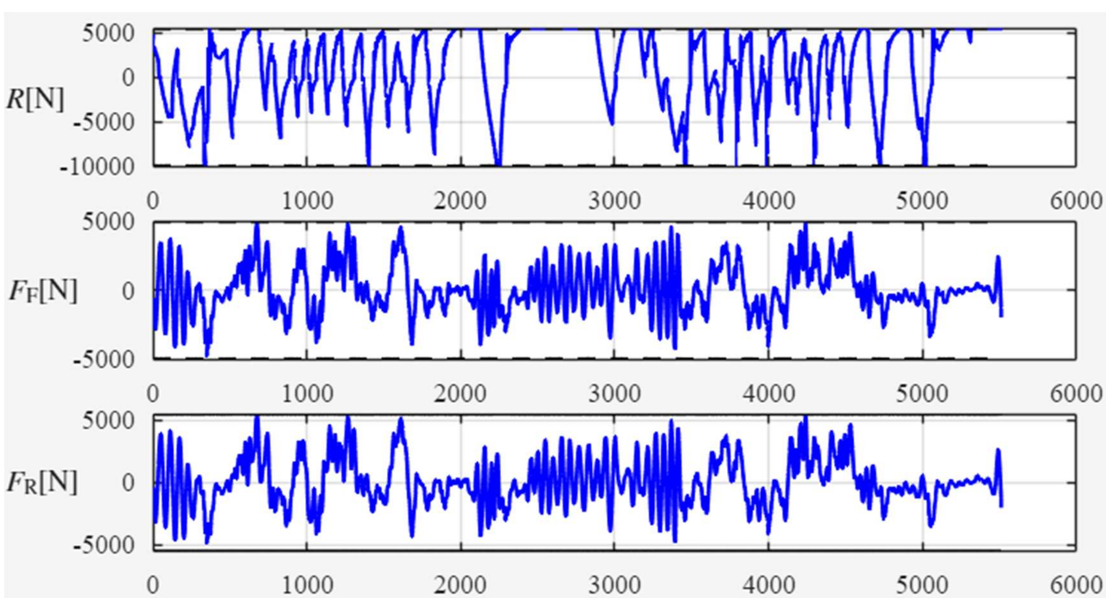

Figure 6: Force Plots of Closed-Loop Simulation

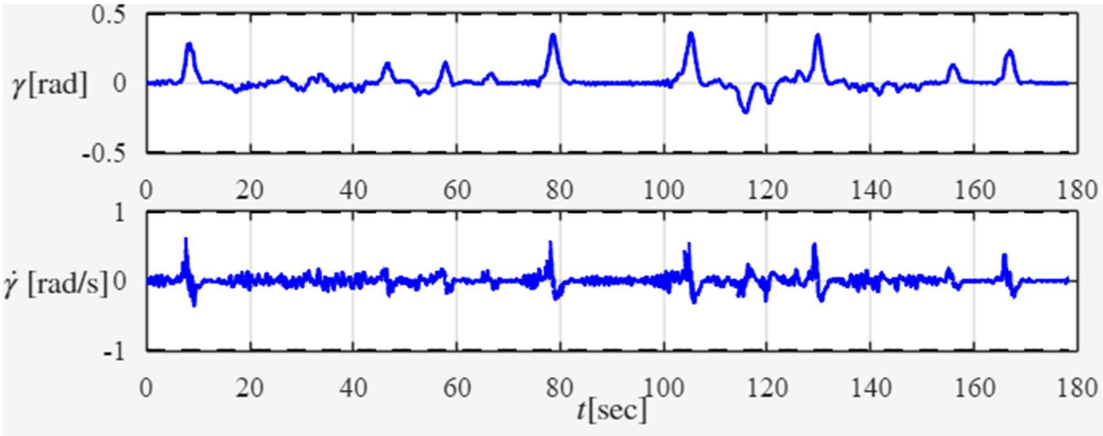

Figure 7: Steering Angle & Rate of Closed-Loop Simulation

The combination of these synchronized inputs results in a zero-violation lap with the spatial path shown below in Figure 8. Ultimately, the closed-loop simulation confirms the viability of the vehicle's performance within its designated constraints, validating the robustness of the SIL architecture, the tuned discrete time controllers, and the optimized racing trajectory.

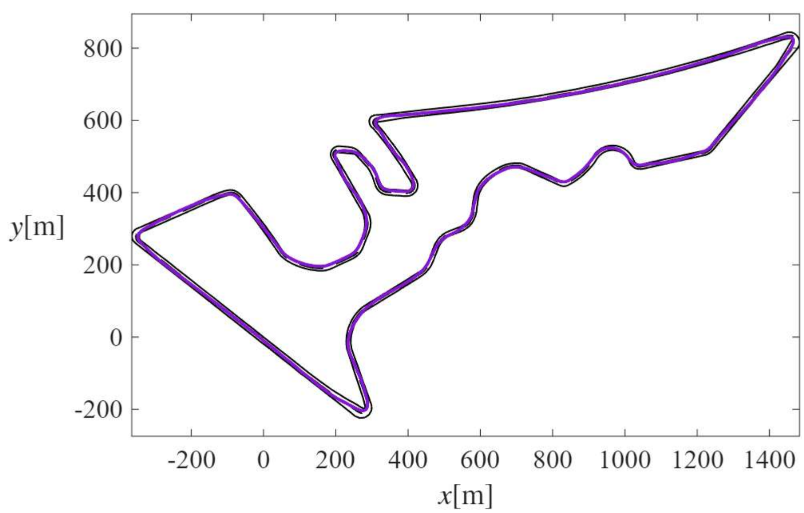

Figure 8: Closed-Loop Simulation Path Plot

This project successfully developed a decoupled SIL architecture that could navigate a highly constrained track in 178.30 seconds. Operating within real-world F1 car parameters, the system achieved zero force or track boundary violations. Beyond just the system's performance, the project yielded key engineering insights. Particularly, the project demonstrated that a controller is only as good as its target, which must also be highly optimized. It also displayed the major tradeoffs and considerations that need to be made when pushing a vehicle's performance to its limits.

Importantly, the analysis of the open-loop state divergence demonstrated that in vehicle dynamics simulation, the mathematical environment (ODE solvers, data extraction methods) is just as important as the control algorithms themselves. Future iterations of the system's architecture could include replacing the bicycle model with a multi-body dynamic model that accounts for vehicle roll and weight transfer. Furthermore, they could also include the upgrading of the LQR steering control to a Model Predictive Controller (MPC) that permits the vehicle to find its own look-ahead trajectory dynamically instead of relying on a pre-generated path.